

Ciem Cornelissen

PhD Researcher in Computer Science Engineering

ciemcornelissen.github.io | ciem.cornelissen@ugent.be | linkedin.com/in/ciem-cornelissen | github.com/ciemcornelissen

ORCID: 0009-0003-0149-6722 | Ghent, Belgium

EDUCATION

Ghent University - imec

PhD in Computer Science Engineering – Deep learning & Robotic sensing

- Affiliated with **IDLab**, Department of Information Technology

Ghent, Belgium

Jun. 2024 – Present

KU Leuven

Advanced Master of Artificial Intelligence – Magna Cum Laude

- Specialization: Deep Learning, Computer Vision
- Master's Thesis: "Quantum Computing for Earth Observation"

Leuven, Belgium

Sep. 2022 – Jun. 2023

Ghent University

Master of Science in Physics – Cum Laude

- Master's Thesis: "Criticality and Forecasting of the Cryptocurrency Market"

Ghent, Belgium

Sep. 2020 – Jun. 2022

RESEARCH EXPERIENCE

PhD Researcher

IDLab, Ghent University - imec

- Developed LISA (Light-Invariant Spectral Autoencoder), a domain-adversarial deep learning framework for illumination-invariant hyperspectral imaging analysis, improving quality prediction generalization by over 20%
- Created the OHSLIC algorithm for efficient UAV-based hyperspectral segmentation, achieving a 10x reduction in inference time (12.2ms vs 130.2ms) and a 25% improvement in prediction accuracy (R^2) over previous methods on an NVIDIA Jetson Nano.
- Built end-to-end IoT-enabled robotic system for non-destructive grape yield and quality mapping in precision viticulture, achieving 0.82 recall for bunch detection and $R^2 = 0.76$ for weight prediction

Jun. 2024 – Present

Ghent, Belgium

PUBLICATIONS

Le MuMo JEPA: Multi-Modal Self-Supervised Representation Learning

2026

- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) – URVIS Workshop* (Jun. 2026)
- Self-supervised framework learning unified representations from RGB and companion modalities via learnable fusion tokens
- Strongest performance-efficiency trade-off on downstream tasks among from-scratch multimodal baselines
- Authors: **C. Cornelissen**, S. Leroux, P. Simoens

In-Field Mapping of Grape Yield and Quality with Illumination-Invariant Deep Learning

2025

- IEEE Internet of Things Journal* (Oct. 2025)
- Developed LISA framework for illumination-invariant hyperspectral imaging in precision viticulture
- Complete IoT-enabled robotic system for non-destructive, real-time mapping of grape yield and quality
- Authors: **C. Cornelissen**, S. De Coninck, A. Willekens, S. Leroux, P. Simoens

Adaptive Clustering for Efficient Phenotype Segmentation of UAV Hyperspectral Data

2025

- IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) Workshops* (Mar. 2025)
- Introduced OHSLIC algorithm for computationally efficient, real-time phenotype segmentation
- Enables on-device processing on UAVs with superior accuracy and reduced inference time
- Authors: **C. Cornelissen**, S. Leroux, P. Simoens

TECHNICAL SKILLS

Machine Learning & AI: Deep Learning, Computer Vision, Domain-Adversarial Learning, Adaptive Neural Networks, Reinforcement Learning, PyTorch, TensorFlow

Sensors & Data Processing: Hyperspectral Imaging (HSI), RGB and SWIR Sensors, Multi-modal Sensor Fusion, UAV-based Remote Sensing, Spectral Analysis

Programming Languages: Python, C/C++, Java, Matlab, Prolog, $\text{\LaTeX}$

Software Development: Git, Docker, IoT Systems, Field Robotics, Linux

Languages: Dutch (Native), English (Fluent, C2 level)